

Area of a circle



1

- | | | | |
|---|--|---|---|
| <p>a</p> <p>i $64\pi \text{ cm}^2$</p> <p>c</p> <p>i $25\pi \text{ cm}^2$</p> <p>e</p> <p>i $324\pi \text{ cm}^2$</p> | <p>ii 201.06 cm^2</p> <p>ii 78.54 cm^2</p> <p>ii 1017.88 cm^2</p> | <p>b</p> <p>i $49\pi \text{ cm}^2$</p> <p>d</p> <p>i $\frac{169\pi}{4} \text{ cm}^2$</p> <p>f</p> <p>i $16\pi \text{ cm}^2$</p> | <p>ii 153.94 cm^2</p> <p>ii 132.73 cm^2</p> <p>ii 50.27 cm^2</p> |
|---|--|---|---|

2

- | | | |
|-------------------------------|-------------------------------|--------------------------------|
| a $16\pi \text{ mm}^2$ | b $16\pi \text{ mm}^2$ | c $144\pi \text{ cm}^2$ |
|-------------------------------|-------------------------------|--------------------------------|

3

- | | | |
|--------------------------------|--------------------------------|--------------------------------|
| a 254.47 cm^2 | b 452.39 cm^2 | c 452.39 cm^2 |
|--------------------------------|--------------------------------|--------------------------------|

4

- | | | | |
|--|--|--|--|
| <p>a</p> <p>i 6 mm</p> <p>c</p> <p>i $\frac{9}{\pi} \text{ cm}$</p> | <p>ii $36\pi \text{ mm}^2$</p> <p>ii $\frac{81}{\pi} \text{ cm}^2$</p> | <p>b</p> <p>i 7 cm</p> <p>d</p> <p>i 11 mm</p> | <p>ii $49\pi \text{ cm}^2$</p> <p>ii $121\pi \text{ cm}^2$</p> |
|--|--|--|--|

5

- | | | | |
|--|--|-----------------------------|-----------------------------|
| a $r = 8 \text{ cm}$ | b $r = \frac{6}{\sqrt{\pi}} \text{ cm}$ | c $r = 6 \text{ cm}$ | d $r = 4 \text{ cm}$ |
| e $r = \frac{5}{\sqrt{\pi}} \text{ mm}$ | | | |

6

- | | |
|---------------------------------|----------------------------------|
| a | b |
| i 5 cm ii $10\pi \text{ cm}$ | i 11 cm ii $22\pi \text{ cm}$ |

7

- | | |
|----------------------|------------------|
| a $r = 10.59$ | b 66.5 cm |
|----------------------|------------------|

8

- | | |
|---|------------------|
| a $\frac{12}{\sqrt{\pi}} \text{ mm}$ | b 6.77 mm |
|---|------------------|

9

- | | |
|---------------|---------------|
| a 4 cm | b 8 cm |
|---------------|---------------|



10

a $\frac{5}{\sqrt{\pi}} \text{ mm}$

b $\frac{10}{\sqrt{\pi}} \text{ mm}$

11

a $\frac{24}{\sqrt{\pi}} \text{ mm}$

b 13.54 mm

12

a 14.1 cm²

b 19.6 mm²

c 95.0 m²

d 115.5 cm²

Applications

13 314.16 cm²

14

a 40 m

b 251.33 m

c 5026.55 m²

15

a 399 832.26 cm²

b 399 629.57 cm²

c Yes, because the approximate area is 202.69 cm² smaller than the actual area.